

Virtual Sensor-Guided Noise Diagnosis and Self-Correction Framework for Sensor Health Monitoring

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Digital Engineering/ Virtual Sensor

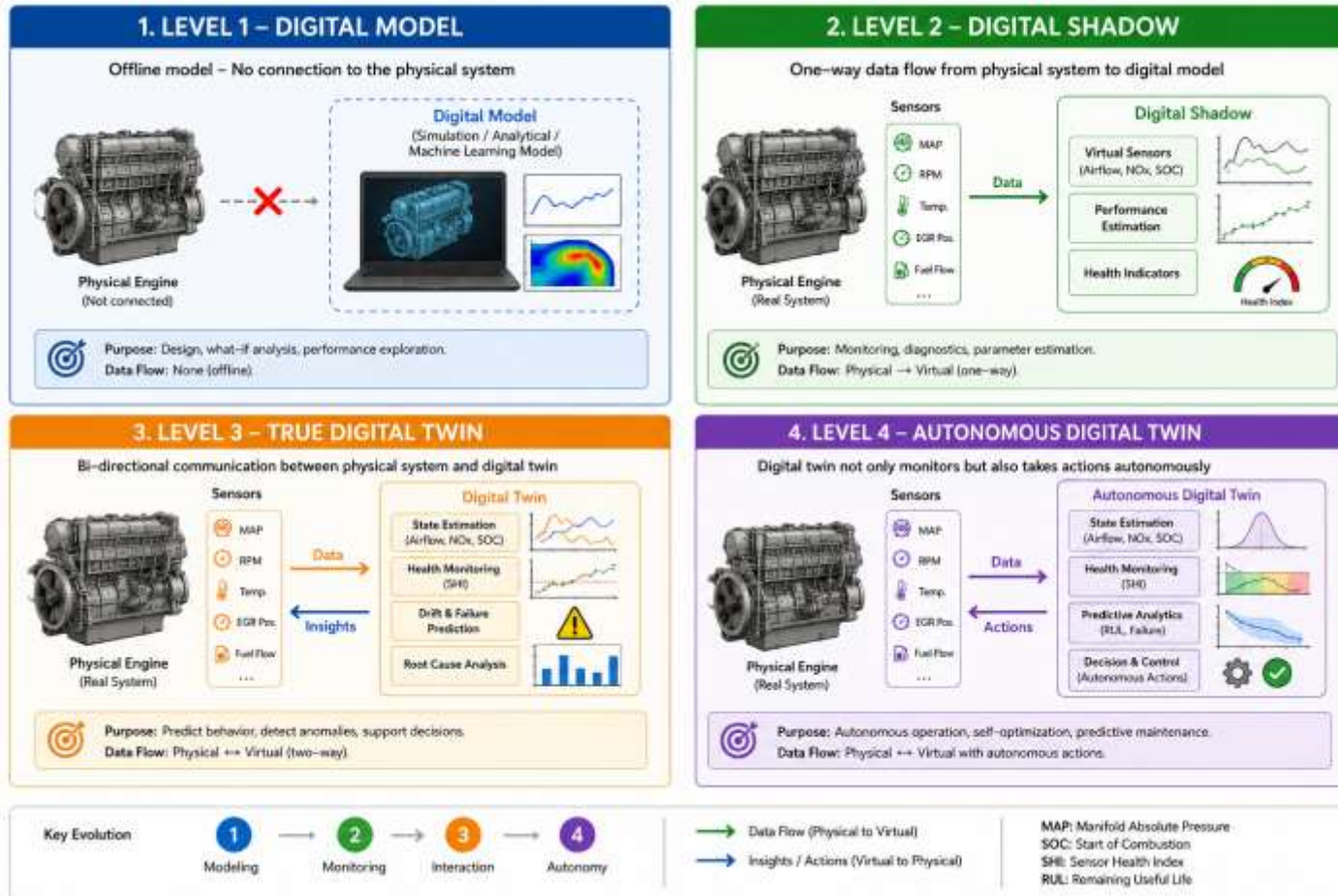


Image Courtesy: Generated using GPT Date: 6/8/2026

Virtual Sensor Principle:

Model-based estimation of difficult-to-measure or important variables

- **Physics perspective**
ECU signals indirectly encode
 - Air-handling dynamics
 - Fuel delivery and engine load
 - Thermodynamic boundary conditions

- **Key advantage**
Information recovery without additional hardware sensors

Virtual sensing shifts the challenge of hardware reliability into a robust, cost-effective software estimation solution.

Virtual Sensor Model Framework

Experimental dataset

- 12-L six-cylinder heavy-duty diesel engine
- 217 steady-state operating points (80% training, 20% testing)
- Wide range of speed and load conditions

Model inputs (ECU signals)

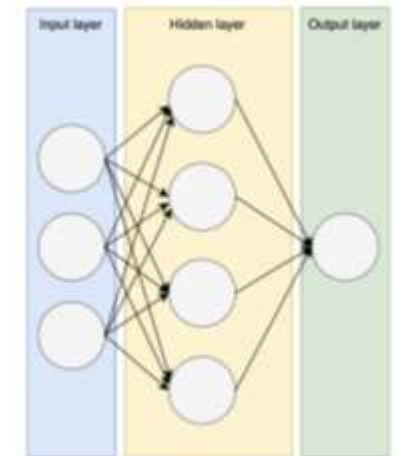
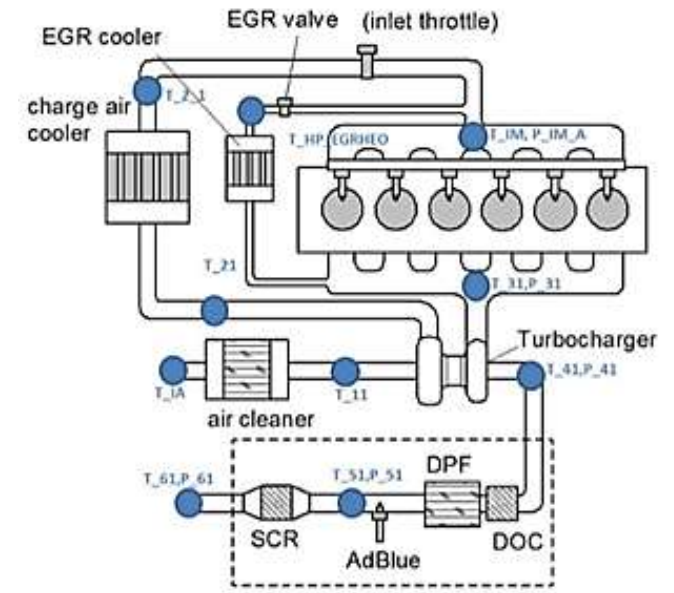
- Engine torque
- Ambient pressure
- Intake manifold temperature
- Intake manifold pressure
- EGR rate
- Variable turbine geometry position

Single Output

- MF_IA

Multi-target

- MF_IA
- NO_x
- SOC



Neural Network Architecture

Proposed model: Artificial Neural Network

- Structure: $6 \xrightarrow{W^{[3]}} 64 \xrightarrow{W^{[4]}} 32 \xrightarrow{W^{[5]}} 16 \xrightarrow{W^{[6]}} 3$
- nonlinear feature learning
- shared representation for multi-target prediction
- simultaneous prediction of three combustion variables

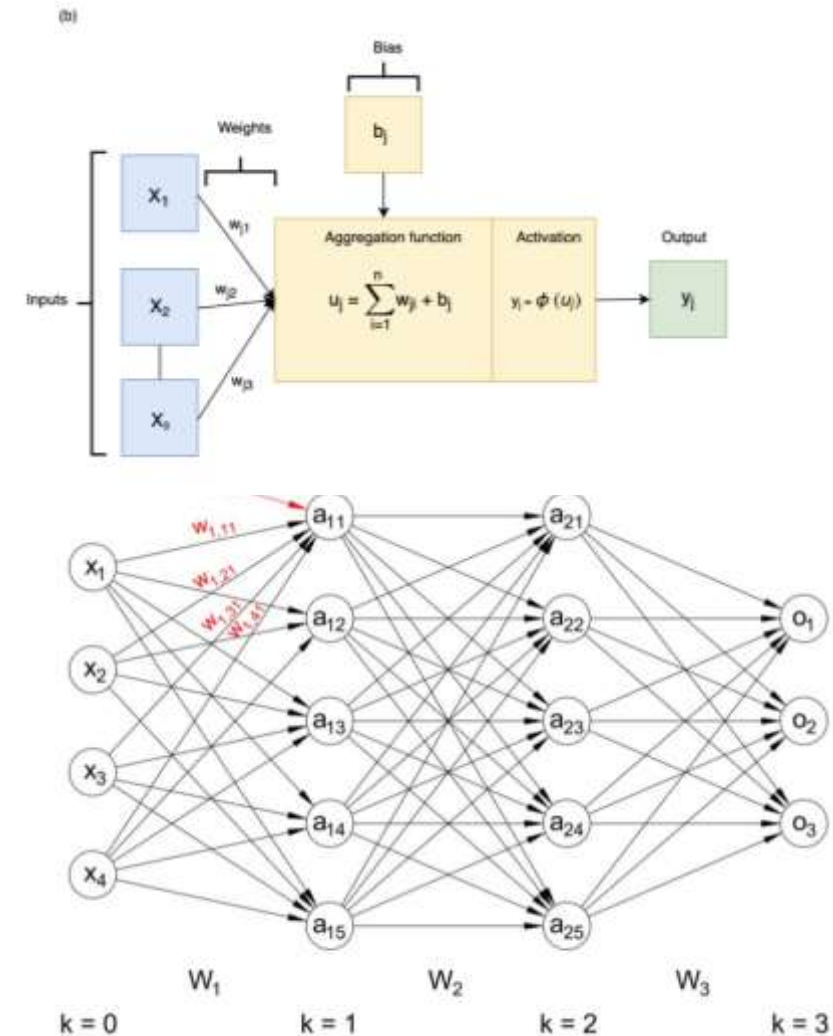
Activation:

- ReLU

Baseline Model

- OLS
 - linear regression reference
- Random Forest
 - tree ensemble capturing nonlinear interactions
- Support Vector Regression
 - kernel-based nonlinear regression

The proposed multi-layer feedforward network enables simultaneous multi-target prediction, outperforming traditional linear and machine learning baselines.



Source: A Gentle Introduction to Physics-Informed Neural Networks, with Applications in Static Rod and Beam Problems - Scientific Figure on ResearchGate. Available from: https://www.researchgate.net/figure/ANN-with-4-inputs-2-hidden-layers-with-5-neurons-and-3-outputs-Weights-of-connections_fig1_361148205 [accessed 9 Mar 2026]

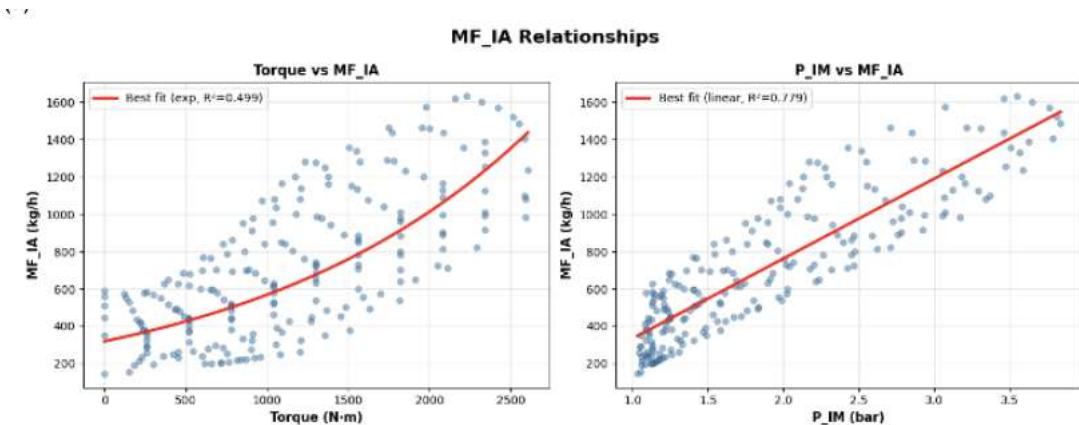
Physical coupling between combustion variables

Individual Physics Pathways:

- *Airflow (MF_{IA}):*
 - Direct determinant of oxygen availability in the cylinder.
- *EGR (Exhaust Gas Recirculation) Rate:*
 - Dilutes oxygen concentration, effectively reducing the peak combustion temperature.
- *SOC (Start of Combustion):*
 - Controls the heat release phasing of the engine cycle.
- *Emission (NO_x Formation):*
 - Highly sensitive to local in-cylinder temperature and oxygen concentration.

Inter-Correlation Between Outputs: (The Thermodynamic Chain)

- Airflow $\rightarrow$ directly influences SOC .
- SOC $\rightarrow$ directly shapes the pressure and temperature fields.
- Temperature + Oxygen concentration $\rightarrow$ ultimately govern NO_x formation.

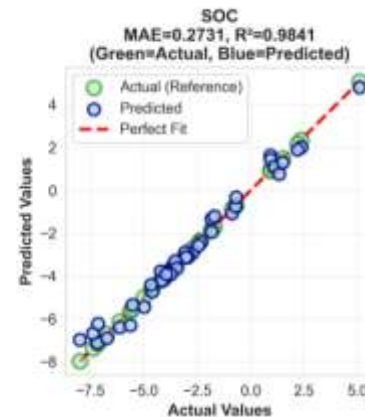
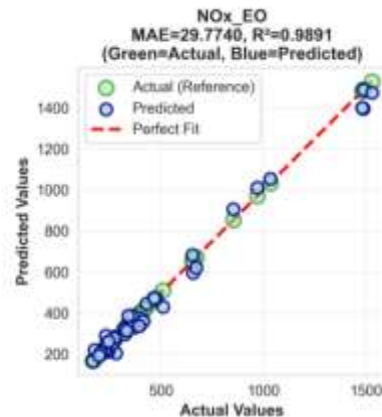
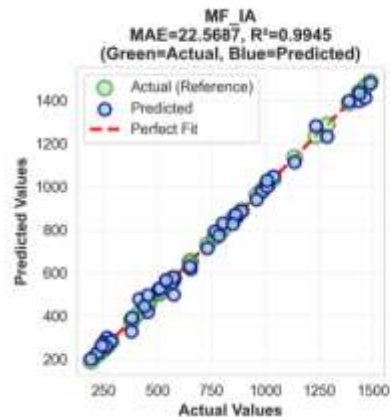


The close thermodynamic inter-correlations among airflow, SOC, and emissions physically justify the use of a shared-representation neural network.

Virtual Sensor Performance

Target	Metric	OLS	Random Forest	SVR (RBF)	ANN (6 inputs)
MF_IA	MAE	49.9	40.4	162.0	22.6
	R ²	0.973	0.981	0.647	0.995
NOx_EO	MAE	89.9	37.5	156.7	29.8
	R ²	0.878	0.974	0.244	0.989
SOC	MAE	1.30	0.40	0.22	0.27
	R ²	0.680	0.958	0.984	0.984

➤ The proposed multi-task ANN outperforms all baseline models, achieving exceptional prediction accuracy ($R^2 > 0.98$) across all three targets.



➤ Clearly Demonstrates, a physics-based selection of the model results in the good accuracy.

Sensitivity Analysis

Target	Input	Baseline MAE	Δ MAE (permute input)	Relative importance $I_{j,k}$
MF_IA	P_IM	22.57	181.87	1.00
	Torque		116.64	0.64
	p_0		108.14	0.59
	T_IM		83.39	0.46
	ECU_VTG_Pos		68.82	0.38
	EGR_Rate		47.36	0.26
Nox_EO	EGR_Rate	29.77	203.33	1.00
	T_IM		111.53	0.55
	Torque		58.92	0.29
	P_IM		39.86	0.20
	p_0		32.13	0.16
	ECU_VTG_Pos		26.45	0.13
SOC	Torque	0.273	1.878	1.00
	P_IM		0.763	0.41
	T_IM		0.741	0.39
	p_0		0.469	0.25
	EGR_Rate		0.458	0.24
	ECU_VTG_Pos		0.354	0.19

- Physical Consistency Established:**

- P_{IM} acts as the primary driver for MF_{IA}
- EGR Rate** dominates NO_x prediction
- Torque** governs SOC

This clear hierarchy proves that a single sensor's failure will propagate uniquely through our multi-task neural network.

The neural network successfully maps thermodynamic relationships, establishing a distinct sensitivity hierarchy where specific sensor failures will dictate overall estimation health.

Representative Case Selection & Noise Injection

Quantile-Median Based Representative Point Selection

Data Segmentation

- Split the complete engine dataset into Low ($\leq 10\%$) and High ($\geq 90\%$) boundary groups
- Isolate extreme operating conditions

Target Selection Criteria

- Calculate the median value within each extreme boundary group
- Establish stable physical baseline, shielding the representative cases from transient outliers.

Point Identification

- Extract the exact time-series sample index t that minimizes the absolute distance to the group median
- $t_{\text{representative}} = \arg \min_t |y(t) - M_{\text{group}}|$

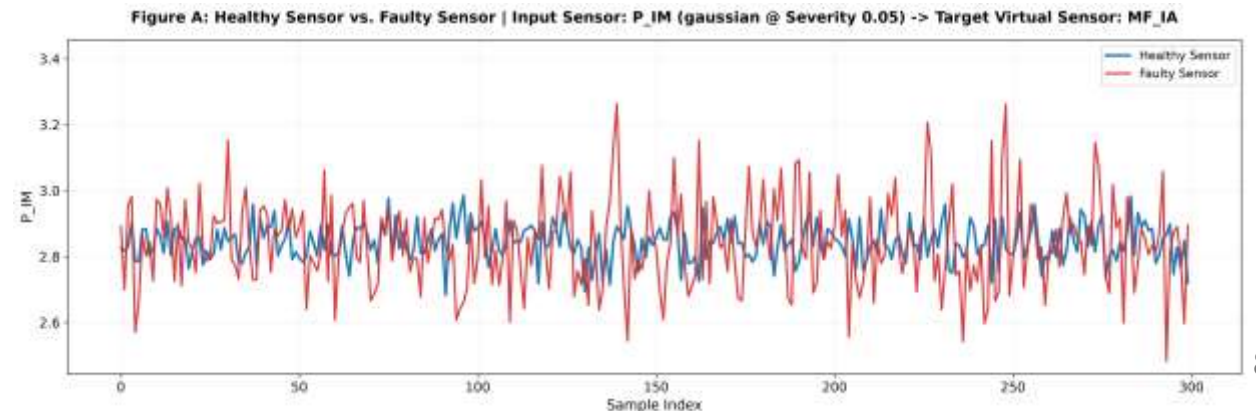
Noise Injection (Gaussian Noise for MF_IA_High)

Healthy Sensor Input (y_{healthy} with 2% noise level):

$$y_{\text{healthy}}(t) = y_{\text{pure}} + \eta_{\text{healthy}}(t)$$
$$\eta_{\text{healthy}}(t) \sim \mathcal{N}(0, \sigma_{\text{healthy}}^2)$$

Faulty Sensor Input (y_{faulty} with 5% severity, $\theta=0.05$):

$$y_{\text{faulty}}(t) = y_{\text{pure}} + \eta_{\text{faulty}}(t)$$
$$\eta_{\text{faulty}}(t) \sim \mathcal{N}(0, \sigma_{\text{faulty}}^2)$$



Noise Diagnosis Results

Figure B: Residual Diagnostics with Semantic Severity Bands | Input Sensor: P_IM (gaussian @ Severity 0.05) -> Target Virtual Sensor: MF_IA

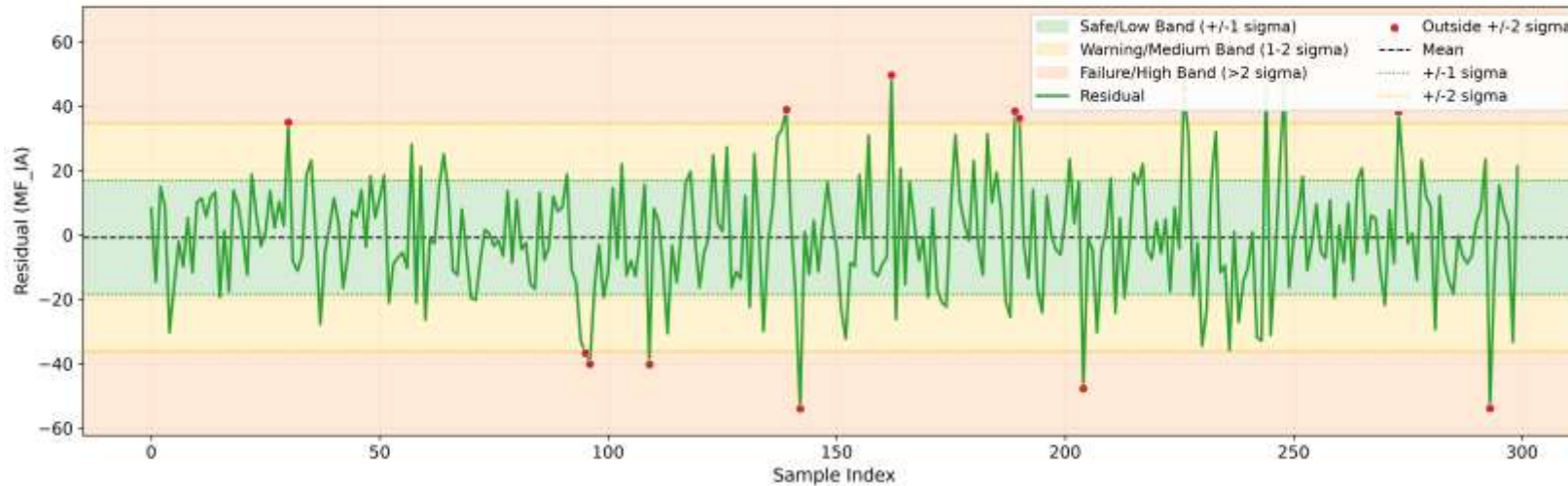
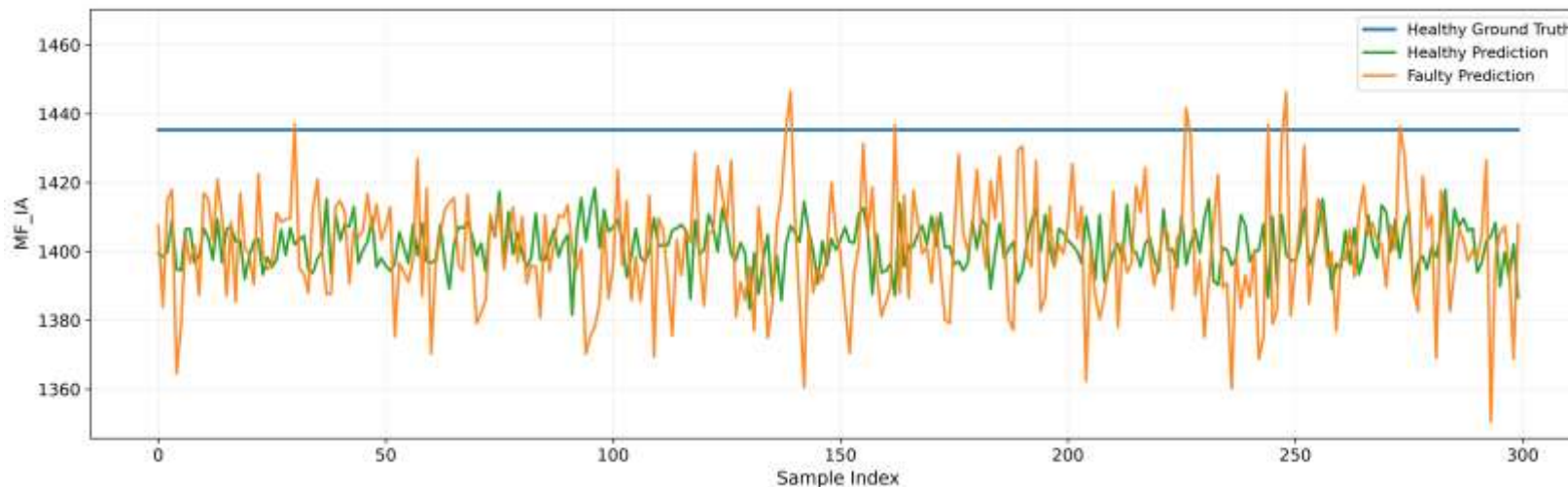


Figure C: Prediction Performance Degradation Path | Input Sensor: P_IM (gaussian @ Severity 0.05) -> Target Virtual Sensor: MF_IA



Noise in Residual Space:

- **Observation:** The residual shows severe high-frequency fluctuations and spikes.
- **Mechanism:** Since healthy and faulty inputs are driven by independent noise seeds (η_{healthy} vs. η_{faulty}), the noise does not cancel out.

Prediction performance:

- **Observation:** Healthy and faulty predictions drift downward to ~ 1400 kg/h.
- **Mechanism:** High-frequency input noise passing through the non-linear neural network causes a systematic negative shift in output expectation, illustrating Systemic Rectification Bias.

Future Work

